

KOTELNIKOVO, Russian Federation

WMO: 346550

Lat: 47.633N

Lon: 43.150E

Elev: 47

StdP: 100.76

Time Zone: 4.00 (E04)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-20.1	-16.8	-23.0	0.5	-20.0	-19.6	0.7	-16.1	10.7	-2.1	9.4	-0.1	1.8	70	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	34.6	19.4	33.0	18.9	31.3	18.7	21.9	30.0	21.0	28.7	20.2	27.6	3.9	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.3	24.3	18.5	13.5	23.7	17.6	12.7	23.3	64.1	30.2	61.0	28.5	58.2	27.7	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.8	7.0	-21.8	37.5	3.8	1.8	-24.5	38.8	-26.7	39.8	-28.8	40.8	-31.5	42.1		
(5)			-22.0	23.1	3.7	2.2	-24.7	24.7	-26.8	26.1	-28.9	27.3	-31.6	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-4.0	-4.0	1.8	10.1	16.8	21.7	23.6	22.8	16.5	9.3	1.6	-2.1		
	DBStd	11.20	5.92	6.00	5.04	4.62	4.09	3.48	2.67	3.29	3.96	4.82	5.54	5.38		
	HDD10.0	1841	433	392	256	56	3	0	0	0	3	71	253	375		
	HDD18.3	3666	691	625	514	248	78	9	0	4	81	282	501	633		
	CDD10.0	1691	0	0	1	58	214	350	423	397	197	49	2	0		
	CDD18.3	475	0	0	0	1	31	109	164	143	25	2	0	0		
	CDH23.3	5189	0	0	0	17	389	1111	1775	1602	273	23	0	0		
	CDH26.7	2118	0	0	0	1	120	464	767	698	67	1	0	0		
(14) Wind	WSAvg	3.2	3.8	3.5	3.8	3.3	2.9	2.9	2.5	2.6	2.8	2.9	3.4	3.8		
(15) Precipitation	PrecAvg	364	31	17	26	27	38	42	31	29	29	26	29	32		
	PrecMax	488	61	34	70	56	76	92	70	62	99	101	67	77		
	PrecMin	218	5	5	3	5	8	12	9	6	3	4	6	11		
	PrecStd	64	14	7	18	16	23	25	21	15	23	22	21	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	9.7	16.2	25.7	32.9	35.7	36.9	37.0	30.9	26.0	15.9	10.1		
		MCWB	7.2	7.6	10.0	13.1	16.1	18.9	21.4	19.1	17.4	16.1	11.7	8.6		
	2%	DB	5.3	6.0	13.1	22.5	29.6	33.6	34.6	34.4	28.6	22.4	12.8	7.5		
		MCWB	4.3	4.3	7.8	13.1	15.9	19.2	20.4	18.7	17.3	14.8	10.6	6.1		
	5%	DB	3.5	4.0	10.8	20.6	27.3	31.6	32.8	32.6	26.6	18.8	10.5	5.9		
		MCWB	2.7	2.6	6.6	12.3	15.7	18.9	19.2	18.8	16.4	13.4	8.9	4.8		
	10%	DB	2.3	2.4	8.6	18.6	25.1	29.5	31.0	30.6	24.5	16.3	8.8	3.9		
		MCWB	1.6	1.3	5.7	11.6	15.2	18.9	18.8	18.6	15.5	12.1	7.3	3.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	7.4	10.8	15.6	18.6	23.7	23.2	22.2	20.6	17.4	12.8	9.0		
		MCDB	8.0	9.7	14.6	21.6	25.0	29.9	34.7	29.9	27.4	23.4	15.1	10.1		
	2%	WB	4.4	4.4	8.8	14.1	17.7	22.0	21.7	21.1	18.6	15.4	10.6	6.3		
		MCDB	5.1	5.7	12.3	20.2	24.5	29.3	30.5	29.2	25.5	20.4	12.3	7.4		
	5%	WB	2.9	2.8	7.1	13.0	16.9	20.8	21.0	20.3	17.6	14.1	9.0	4.8		
		MCDB	3.4	3.8	9.9	18.3	24.4	27.9	28.9	28.3	23.7	18.1	10.3	5.8		
	10%	WB	1.6	1.4	5.8	12.0	16.1	19.7	20.2	19.5	16.5	12.7	7.4	3.2		
		MCDB	2.2	2.2	8.6	17.8	23.5	26.4	27.6	27.5	22.3	16.2	8.8	3.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	6.0	7.4	10.9	11.8	11.3	11.9	12.4	11.7	10.0	6.4	4.9		
		MCDBR	5.2	6.5	11.9	14.8	15.4	14.9	14.9	15.8	15.4	14.4	8.2	5.5		
		MCWBR	4.4	5.1	7.5	7.4	5.5	5.4	5.1	5.3	7.0	7.9	6.2	4.6		
		MCDBR	5.2	5.5	10.6	12.6	12.4	12.2	12.0	12.6	12.2	11.5	7.5	5.4		
	5% WB	MCWBR	4.7	4.6	7.1	7.0	5.5	5.8	4.6	5.4	6.3	6.7	5.9	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.341	0.374	0.408	0.455	0.426	0.420	0.441	0.443	0.416	0.381	0.353	0.344		
	taud		2.162	2.083	2.115	2.043	2.219	2.282	2.229	2.200	2.233	2.285	2.266	2.180		
	Ebn at Noon		697	754	796	795	840	846	821	799	782	748	685	639		
	Edh at Noon		87	116	132	156	136	129	134	132	117	95	78	76		
(44) All-Sky Solar Radiation	RadAvg		0.91	1.73	2.94	4.64	5.95	6.50	6.37	5.68	4.14	2.40	1.11	0.67		
	RadStd		0.22	0.31	0.32	0.43	0.56	0.47	0.30	0.37	0.40	0.34	0.19	0.08		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.53	N/A	N/A	N/A	N/A	N/A	N/A	+135	+80

Nomenclature: See separate page